

NOTES 19: CLT-BASED INFERENCE FOR PROPORTIONS

Stat 120 | Spring 2026

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CLT: The Central Limit Theorem (CLT) tells us that if the sample size is big enough and the sample is random,

$$\bar{X} \sim N(\text{____}, \text{____})$$

$$\hat{p} \sim N(\text{____}, \text{____})$$

The general form for a **confidence interval** is:

Example: Finding z^*

- 95% confidence interval
- 68% confidence interval
- 99% confidence interval
- 90% confidence interval

0.1 How big is big enough?

Example 1: $\hat{p} = .5$

Example 1: $\hat{p} = .05$

Rule of Thumb for Proportions:

- Expected count in each category (Yes/No) should be $> \text{____}$
- $np > \text{____}$ and $n(1 - p) > \text{____}$

0.2 How do we find the SE?

Standard Error for Proportions:

Idea: As n gets bigger, the SE gets _____.

If $\hat{p}$ is close to .5, the SE is _____ than if $\hat{p}$ is close to 0 or 1

Example: ESP Example Again

$n = 14$

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p_hat = 3/14
n = 14
SE_p = sqrt((p_hat*(1-p_hat))/n)
z_score = (p_hat - .2)/SE_p
p_val = pnorm(z_score, lower.tail = FALSE)
p_val
```

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[1] 0.4482
```

$n=1400$:

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Big Picture Picture

1 In-class Group Questions

1. APM Research Lab ran a [survey of likely Minnesota voters](#) between Sept 16-18 of 2024. They reported 48.4% in support of the Harris/Walz ticket, and 43.3% in support of the Trump/Vance ticket.

Here's an excerpt from their methodology report:

The margin for error, according to standards customarily used by statisticians, is no more than ± 3.5 percentage points. This means that there is a 95% probability that the "true" figure would fall within that range if all voters were surveyed.

- (a) Show how the margin of error was computed
 - (b) Is it OK to use the normal distribution as a model for the sampling distribution of a proportion in this case?
2. In a random sample of 100 moviegoers in January 2013, 64 of them said they are more likely to wait and watch a new movie in the comfort of their own home.
 - (a) Is it ok to use the normal distribution as a model for the sampling distribution of the proportion of moviegoers preferring to wait and watch at home?
 - (b) Suppose I want to know if more than 50% of moviegoers in the population prefer to wait and watch at home. Write down the corresponding null and alternative hypotheses.
 - (c) For the hypothesis test in part (b), find the standardized test statistic (i.e., the Z-score).

(d) Calculate the p-value and state a conclusion in context

(e) The table below gives some of the percentiles of the $N(0,1)$ distribution. If you were given this table (for example on an exam, where you do not have access to R), what could you say about the value of the p-value?

percentage	percentile (qnorm(percentage))
90%	1.3
95%	1.6
97.5%	2.0
99%	2.3
99.5%	2.6

(f) Compute and interpret a 90% confidence interval for the population proportion of moviegoers who believe they are more likely to watch a new movie from home. (Use the information from the previous table to find the critical value needed.)

(g) How big of a sample size would you need if you wanted the margin of error for a 90% confidence interval to be no more than 2%? To answer this question, assume that the true proportion is $p = 0.5$, which would result in the biggest margin of error for a given sample size.